JAVA SCRIPT

CHAPTER-5

5.1 CONDITIONAL STATEMENTS

Very often when you write code, you want to perform different actions for different decisions. You can use conditional statements in your code to do this.
In JavaScript we have the following conditional statements:

- Use if to specify a block of code to be executed, if a specified condition is true
- Use else to specify a block of code to be executed, if the same condition is false
- Use else if to specify a new condition to test, if the first condition is false
- Use switch to specify many alternative blocks of code to be executed

The if Statement

Use if statement to specify a block of JavaScript code to be executed if a condition is true.

Syntax

```
if (condition) {
  //  block of code to be executed if the condition is true
}
```

PROGRAM 1:

```
<!DOCTYPE html>
<html>
<body>
<h2>JavaScript if</h2>
<p>Display "Good day!" if the hour is less than 18:00:</p>
<p id="demo">Good Evening!</p>
<script>
if (new Date().getHours() < 18) {
  document.getElementById("demo").innerHTML = "Good day!";
}
</script>
</body>
</html>
```

The else Statement

Use the else statement to specify a block of code to be executed if the condition is false.

Syntax:

```
if (condition) {
  // block of code to be executed if the condition is true
} else {
  // block of code to be executed if the condition is false
}
```

PROGRAM 2:

```
<!DOCTYPE html>
<html>
<body>
<h2>JavaScript if .. else</h2>
<p id="demo"></p>
<script>
const hour = new Date().getHours();
let greeting;
if (hour < 18) {
  greeting = "Good day";
} else {
  greeting = "Good evening";
}
document.getElementById("demo").innerHTML = greeting;
</script>
</body>
</html>
```

Else if Statement

Use the else if statement to specify a new condition if the first condition is false.

Syntax

```
if (condition1) {
  // block of code to be executed if condition1 is true
} else if (condition2) {
  // block of code to be executed if the condition1 is false and condition2 is true
} else {
  // block of code to be executed if the condition1 is false and condition2 is false
}
```

PROGRAM 3:

```
<!DOCTYPE html>
<html>
<body>
<h2>JavaScript if .. else</h2>
<p id="demo"></p>
<script>
const time = new Date().getHours();
```

```
let greeting;
if (time < 10) {
  greeting = "Good morning";
} else if (time < 20) {
  greeting = "Good day";
} else {
  greeting = "Good evening";
}
document.getElementById("demo").innerHTML = greeting;
</script>
</body>
</html>
```

Switch Statement

Use the switch statement to select one of many code blocks to be executed.

Syntax

```
switch(expression) {
  case x:
    // code block
    break;
  case y:
    // code block
    break;
  default:
    // code block
}
```

- The switch expression is evaluated once.
- The value of the expression is compared with the values of each case.
- If there is a match, the associated block of code is executed.
- If there is no match, the default code block is executed.

PROGRAM 4:

```
<!DOCTYPE html>
<html>
<body>
<h2>JavaScript switch</h2>
<p id="demo"></p>
<script>
let day;
switch (new Date().getDay()) {
  case 0:
    day = "Sunday";
```

```
  break;
 case 1:
  day = "Monday";
  break;
 case 2:
  day = "Tuesday";
  break;
 case 3:
  day = "Wednesday";
  break;
 case 4:
  day = "Thursday";
  break;
 case 5:
  day = "Friday";
  break;
 case  6:
  day = "Saturday";
}
document.getElementById("demo").innerHTML = "Today is " + day;
</script>
</body>
</html>
```

5.2 Loops

- for - loops through a block of code a number of times
- for/in - loops through the properties of an object
- for/of - loops through the values of an iterable object
- while - loops through a block of code while a specified condition is true
- do/while - also loops through a block of code while a specified condition is true

The For Loop

The for statement creates a loop with 3 optional expressions:

Syntax

for (*expression 1*; *expression 2*; *expression 3*) {
 // code block to be executed
}

PROGRAM 5:

```
<!DOCTYPE html>
<html>
<body>
```

```
<h2>JavaScript For Loop</h2>
<p id="demo"></p>
<script>
let text = "";
for (let i = 0; i < 5; i++) {
  text += "The number is " + i + "<br>";
}
document.getElementById("demo").innerHTML = text;
</script>
</body>
</html>
```

The For In Loop

The JavaScript for in statement loops through the properties of an Object:

Syntax

```
for (key in object) {
  // code block to be executed
}
```

PROGRAM 6:

```
<!DOCTYPE html>
<html>
<body>
<h2>JavaScript For In Loop</h2>
<p>The for in statement loops through the properties of an object:</p>
<p id="demo"></p>
<script>
const person = {fname:"John", lname:"Doe", age:25};
let txt = "";
for (let x in person) {
  txt += person[x] + " ";
  }
document.getElementById("demo").innerHTML = txt;
</script>
</body>
</html>
```

The For Of Loop

The JavaScript for of statement loops through the values of an iterable object. It lets you loop over iterable data structures such as Arrays, Strings, Maps, NodeLists, and more:

Syntax

```
for (variable of iterable) {
  // code block to be executed
}
```

Variable - For every iteration the value of the next property is assigned to the variable. *Variable* can be declared with const, let, or var.
Iterable - An object that has iterable properties.

The While Loop

The while loop loops through a block of code as long as a specified condition is true.

Syntax

```
while (condition) {
  // code block to be executed
}
```

PROGRAM 7:

```
<!DOCTYPE html>
<html>
<body>
<h2>JavaScript While Loop</h2>
<p id="demo"></p>
<script>
let text = "";
let i = 0;
while (i < 10) {
  text += "<br>The number is " + i;
  i++;
}
document.getElementById("demo").innerHTML = text;
</script>
</body>
</html>
```

The Do While Loop

The do while loop is a variant of the while loop. This loop will execute the code block once, before checking if the condition is true, and then it will repeat the loop as long as the condition is true.

Syntax

```
do {
  // code block to be executed
```

```
}
while (condition);
```

PROGRAM 8:

```
<!DOCTYPE html>
<html>
<body>
<h2>JavaScript Do While Loop</h2>
<p id="demo"></p>
<script>
let text = ""
let i = 0;
do {
  text += "<br>The number is " + i;
  i++;
}
while (i < 10);
document.getElementById("demo").innerHTML = text;
</script>
</body>
</html>
```

5.3 THE BREAK STATEMENT

You have already seen the break statement used in an earlier chapter of this tutorial. It was used to "jump out" of a switch() statement.The break statement can also be used to jump out of a loop:

PROGRAM 9:

```
<!DOCTYPE html>
<html>
<body>
<h2>JavaScript Loops</h2>
<p>A loop with a <b>break</b> statement.</p>
<p id="demo"></p>
<script>
let text = "";
for (let i = 0; i < 10; i++) {
  if (i === 3) { break; }
  text += "The number is " + i + "<br>";
}
document.getElementById("demo").innerHTML = text;
</script>
</body>
</html>
```

The Continue Statement

The continue statement breaks one iteration (in the loop), if a specified condition occurs, and continues with the next iteration in the loop.

PROGRAM 10:

```
<!DOCTYPE html>
<html>
<body>
<h2>JavaScript Loops</h2>
<p>A loop with a <b>continue</b> statement.</p>
<p>A loop which will skip the step where i = 3.</p>
<p id="demo"></p>
<script>
let text = "";
for (let i = 0; i < 10; i++) {
  if (i === 3) { continue; }
  text += "The number is " + i + "<br>";
}
document.getElementById("demo").innerHTML = text;
</script>
</body>
</html>
```

5.4 JAVASCRIPT ITERABLES

Iterating is easy to understand. It simply means looping over a sequence of elements. Here are some easy examples:

- Iterating over a String
- Iterating over an Array

Iterating Over a String

You can use a for.of loop to iterate over the elements of a string:

PROGRAM 11:

```
<!DOCTYPE html>
<html>
<body>
<h2>JavaScript Iterables</h2>
<p>Iterate over a String:</p>
<p id="demo"></p>
<script>
// Create a String
```

```
const name = "W3Schools";
// List all Elements
let text = ""
for (const x of name) {
  text += x + "<br>";
}
document.getElementById("demo").innerHTML = text;
</script>
</body>
</html>
```

Iterating Over an Array

You can use a for.Of loop to iterate over the elements of an Array:

PROGRAM 12:

```
<!DOCTYPE html>
<html>
<body>
<h2>JavaScript Iterables</h2>
<p>Iterate over an Array:</p>
<p id="demo"></p>
<script>
// Create aa Array
const letters = ["a","b","c"];
// List all Elements
let text = "";
for (const x of letters) {
  text += x + "<br>";
}
document.getElementById("demo").innerHTML = text;
</script>
</body>
</html>
```

Iterating Over a Map

You can use for of loop to iterate over the elements of a Map:

PROGRAM 13:

```
<!DOCTYPE html>
<html>
<body>
<h2>JavaScript Iterables</h2>
<p>Iterate over a Map:</p>
```

```
<p id="demo"></p>
<script>
// Create a Map
const fruits = new Map([
  ["apples", 500],
  ["bananas", 300],
  ["oranges", 200]
]);
// List all entries
let text = "";
for (const x of fruits) {
  text += x + "<br>";
}
document.getElementById("demo").innerHTML = text;
</script>
</body>
</html>
```